INCH-POUND

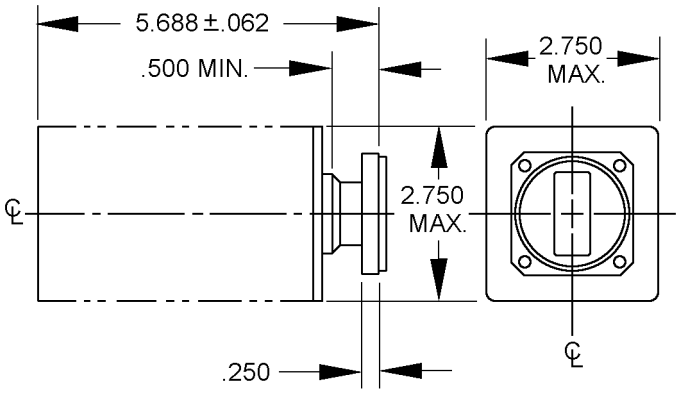
MIL-DTL-3954/19B
w/Amendment 1
5 October 2018
SUPERSEDING
MIL-D-3954/19B
18 February 2015

MILITARY SPECIFICATION SHEET

DUMMY LOADS, ELECTRICAL, WAVEGUIDE (FREQUENCY RANGE 8.2 TO 12.4 GIGAHERTZ)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the product described herein shall consist of this Specification Sheet and MIL-DTL-3954.



Inches	mm
.062	1.57
.250	6.35
.500	12.70
2.750	69.85
5.688	144.48

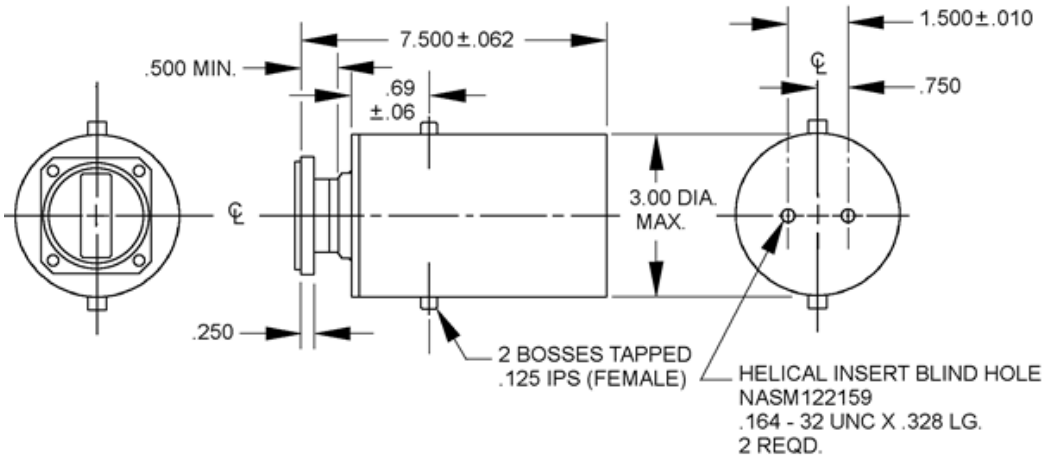
NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only.
4. Screws, lockwashers and gasket are supplied with mating flange.
5. Quantity of fins required will be dependent on heat level requirement of the load.

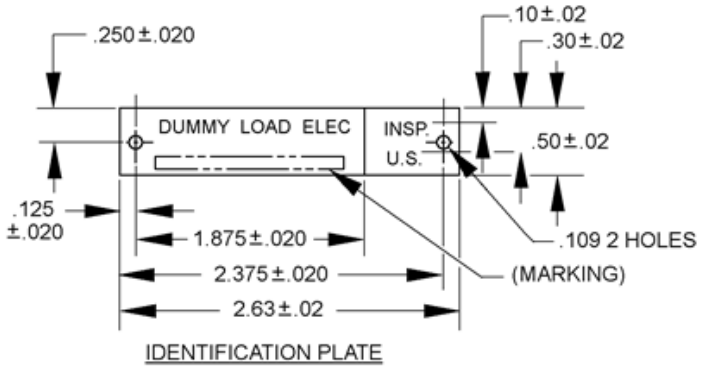
FIGURE 1. Class 1 dummy load.

AMSC N/A

FSC 5985

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Inches	mm	Inches	mm
.010	.25	.328	8.33
.02	.51	.500	12.70
. 05	1.52	.69	17.53
.062	1.57	1.50	38.14
.100	2.54	1.875	47.63
.109	2.77	2.375	60.33
.125	3.18	2.63	66.80
.164	4.17	3.00	76.20
.250	6.35	7.500	189.50
.300	7.62		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only.
4. Screws, lockwashers and gasket are supplied with mating flange.

FIGURE 2. Class II dummy load.

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TABLE I. Characteristics.

PIN	Class	Material	Flange equal to 2/	VSWR max	Power		Pressure		Figure	Flow rate	Input temp.	Output temp.
					Average	Peak	Internal	Coolant chamber				
					(watts)	(kilowatts)	(psig)	(psig)		(GPM 1/)	(°C)	(°C)
M3954/19-01	I	AL	M3922/53-003 (UG-135/U)	1.10	3/ 400	290	30	--	1			
M3954/19-02	I	Copper	M3922/53-001 (UG-39/U)	1.10	400	290	30	100	1			
M3954/19-03	II	AL	M3922/53-003 (UG-135/U)	1.10	1000	290	30	100	2	0.30	35	60
M3954/19-04	II	Copper	M3922/53-001 (UG-39/U)	1.10	1000	290	30	100	2	0.30	35	60
M3954/19-05	II	Corrosion resisting steel	2/M3922/53-001 (UG-39/U)	1.10	1000	290	30	100	2	0.30	35	60

1/ The flow rate was determined from the following formula:

$$Q = \frac{6.8P}{Cp\ \Delta T}$$

Where: Q = Minimum flow rate in GPM
P = Avg power in kilowatts
Cp = Specific heat of coolant.
ΔT = Coolant temperature rise in °F.

The calculations were made for Cp = 1 for water, a ΔT of 45°F and a safety factor of approximately 2. For different coolants of different temperature rises, a different flow rate would be necessary.

2/ Except thickness.

3/ This is a test power value. The average rated power is 500 watts.

TABLE II. Cross reference of PIN/ AN nomenclature.

PIN	AN nomenclature
M3954/19-01	DA-146/U

REQUIREMENTS:

Design and construction

Dimensions and configuration: See figure 1.

Weight:
Dry load – 1.6 lbs max.
Liquid cooled load – 29 lbs max.

Performance characteristics: See table I.

Part or Identifying Number (PIN): M3954/19 – (dash number from table I).

Referenced documents. In addition to MIL-DTL-3954, this document references the following:

NASM122159

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2018-011)

Review activities:
Army – AR, MI
Navy – AS, CG, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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